

Project Title
(Warehouse Inventory
System)

*Project Report submitted for the partial fulfilment of grade for the
course*

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Describe what your project presents within 150 words.

Warehouses manage thousands of products daily, and efficiency depends on fast search, accurate stock tracking, and seamless supplier coordination. This project presents a **Warehouse Inventory System** built with Java, MySQL, and advanced data structures to overcome manual spreadsheet limitations. The system integrates six DSA modules to deliver core workflows: product and supplier search using string algorithms, optimized reordering through dynamic programming, efficient supplier allocation via network flow, quick report generation with approximation techniques, and secure, parallelized product updates. A unified dashboard ensures smooth operations, while algorithms like KMP, RabinKarp, and MaxFlow/MinCut enhance speed and reliability. By combining theoretical algorithms with practical database-driven workflows, the system minimizes delays, prevents low-stock surprises, and boosts warehouse productivity. Ultimately, it provides a **fast, searchable, and always-in-stock solution** for modern inventory management.

Signature of the Guide